

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Epson
Firmware Version: Main: 670078739GWWP410 Main 2: 677325WWP400
Model(s): BrightLink Pro 1460Ui, EB-1460UT, EB-1460Ui, CB-1460Ui, BrightLink 697Ui, EB-1450UT, BrightLink Pro 1450Ui, EB-1450Ui, EB-1440UT, EB-1440Ui, CB-1440Ui, EB-696UT, BrightLink 696Ui, EB-696Ui, CB-696Ui, EB-690U

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.06.0002-b002	1.4.2

Version History

Module Version	Date	Notes
1_0_2_0	1/30/2018	Added Split Screen commands.
1_0_1_0	12/20/2017	Updated power status and Aspect Ratio Status
1_0_0_1	8/31/2017	Update status example
1_0_0_0	8/2/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass

```
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='BrightLink Pro 1460Ui')
```

SerialOverEthernetClass

```
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='BrightLink Pro 1460Ui')
```

EthernetClass

```
InterfaceName = ModuleName.EthernetClass('192.168.254.254', 3629, Model='BrightLink Pro 1460Ui')
```

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value 'Normal' ¹ 'Full'	Value '16:9' 'Zoom'	Value 'Auto' ² 'Real'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Normal')			
Command AutoImage ⁴	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command AVMute	Value 'On'	Value 'Off'	
# AVMute example InterfaceName.Set('AVMute', 'On')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'Input 1 (RGB)'	Value 'Input 1 (Component)'	Value 'HDMI 1'
	'Video (RCA)'	'USB Display'	'USB 1'
	'LAN'	'USB 2'	'HDMI 2'
	'HDMI 3' ⁵	'Whiteboard' ⁶	'Screen Mirroring' ^{3, 6}
# Input example InterfaceName.Set('Input', 'Input 1 (RGB)')			
Command LampMode	Value 'Normal'	Value 'Eco'	
# LampMode example InterfaceName.Set('LampMode', 'Normal')			
Command MenuNavigation	Value 'Menu'	Value 'Escape'	Value 'Enter'
	'Up'	'Down'	'Left'
	'Right'		
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu')			

Command SplitScreen	Value 'On'	Value 'Off'	
# SplitScreen example InterfaceName.Set('SplitScreen', 'On')			
Command SplitScreenLeftInput	Value 'Input 1 (RGB)' 'Video (RCA)' 'LAN' 'HDMI 3' ⁵	Value 'Input 1 (Component)' 'USB Display' 'USB 2' 'Whiteboard' ⁶	Value 'HDMI 1' 'USB 1' 'HDMI 2' 'Screen Mirroring' ^{3, 6}
# SplitScreenLeftInput example InterfaceName.Set('SplitScreenLeftInput', 'Input 1 (RGB)')			
Command SplitScreenMode	Value 'Size 1'	Value 'Size 2'	Value 'Size 3'
# SplitScreenMode example InterfaceName.Set('SplitScreenMode', 'Size 1')			
Command SplitScreenRightInput	Value 'Input 1 (RGB)' 'Video (RCA)' 'LAN' 'HDMI 3' ⁵	Value 'Input 1 (Component)' 'USB Display' 'USB 2' 'Whiteboard' ⁶	Value 'HDMI 1' 'USB 1' 'HDMI 2' 'Screen Mirroring' ^{3, 6}
# SplitScreenRightInput example InterfaceName.Set('SplitScreenRightInput', 'Input 1 (RGB)')			
Command SplitScreenSwap	Value None		
# SplitScreenSwap example InterfaceName.Set('SplitScreenSwap', None)			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command Volume	Value 0 to 20 in steps of 1		
# Volume example InterfaceName.Set('Volume', 20)			

¹ Not supported with inputs: HDMI 1, HDMI 2

² Not supported with inputs: Input 1 (RGB), Input 1 (Component), Video (RCA)

³ Only supported by models: EB-1460UT, EB-1460Ui, CB-1460Ui, BrightLink 697Ui.

⁴ Not supported with inputs: Input 1 (Component), HDMI 1, HDMI 2, Video (RCA)

⁵ Only supported by models: EB-696UT, BrightLink 696Ui, EB-696Ui, CB-696Ui, EB-690U.

⁶ Only supported by models: EB-1450UT, BrightLinkPro 1450Ui, EB-1450Ui, EB-1440UT, EB-1440Ui, CB01440Ui.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
AspectRatio	'Normal' ¹	'16:9'	'Auto' ²
	'Full'	'Zoom'	'Real'
# AspectRatio examples InterfaceName.Update('AspectRatio') value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command	Value	Value	
AVMute	'On'	'Off'	
# AVMute examples InterfaceName.Update('AVMute') value = InterfaceName.ReadStatus('AVMute') InterfaceName.SubscribeStatus('AVMute', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus examples value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	Value
DeviceStatus ⁴	'Normal'	'Fan Error'	'Lamp Out'
	'Internal Temp Error'	'Lamp Error'	'Lamp Cover Error'
# DeviceStatus examples InterfaceName.Update('DeviceStatus') value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)			
Command	Value	Value	
Freeze	'On'	'Off'	
# Freeze examples InterfaceName.Update('Freeze')			

<pre>value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)</pre>			
Command Input	Value 'Input 1 (RGB)' 'Video (RCA)' 'LAN' 'HDMI 3' ⁵	Value 'Input 1 (Component)' 'USB Display' 'USB 2' 'Whiteboard' ⁶	Value 'HDMI 1' 'USB 1' 'HDMI 2' 'Screen Mirroring' ^{3, 6}
<pre># Input examples InterfaceName.Update('Input') value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)</pre>			
Command LampMode	Value 'Normal'	Value 'Eco'	
<pre># LampMode examples InterfaceName.Update('LampMode') value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)</pre>			
Command LampUsage	Value 0 –		
<pre># LampUsage examples InterfaceName.Update('LampUsage') value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)</pre>			
Command Power	Value 'On' 'Cooling Down'	Value 'Off'	Value 'Warming Up'
<pre># Power examples InterfaceName.Update('Power') value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>			
Command Volume	Value 0 to 20 in steps of 1		
<pre># Volume examples InterfaceName.Update('Volume') value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)</pre>			

¹ Not supported with inputs: HDMI 1, HDMI 2

² Not supported with inputs: Input 1 (RGB), Input 1 (Component), Video (RCA)

³ Only supported by models: EB-1460UT, EB-1460Ui, CB-1460Ui, BrightLink 697Ui.

⁴ Device Status is not supported using Ethernet.

⁵ Only supported by models: EB-696UT, BrightLink 696Ui, EB-696Ui, CB-696Ui, EB-690U.

⁶ Only supported by models: EB-1450UT, BrightLinkPro 1450Ui, EB-1450Ui, EB-1440UT, EB-1440Ui, CB01440Ui.

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

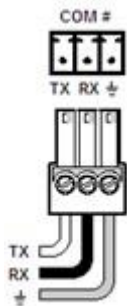
Data Bits: 8

Parity: None

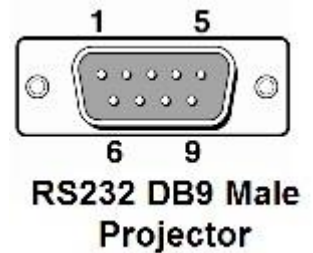
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND	→	5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface

Port Type:	Ethernet
Default Port:	3629
Logon Credentials Supported:	No
Multi-Connection	Yes
Capabilities:	
Port Changeability:	No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

To control over Ethernet, please ensure that Communication On is enabled. To enable Communication On:
Menu -> Extended -> Standby Mode -> Communication On

Appendix A. Set Commands

AV Mute Off	MUTE OFF\x0D
AV Mute On	MUTE ON\x0D
Aspect Ratio 16:9	ASPECT 20\x0D
Aspect Ratio Auto	ASPECT 30\x0D
Aspect Ratio Full	ASPECT 40\x0D
Aspect Ratio Normal	ASPECT 00\x0D
Aspect Ratio Real	ASPECT 60\x0D
Aspect Ratio Zoom	ASPECT 50\x0D
Auto Image None	KEY 4A\x0D
Freeze Off	FREEZE OFF\x0D
Freeze On	FREEZE ON\x0D
Input HDMI 1	SOURCE 30\x0D
Input HDMI 2	SOURCE A0\x0D
Input Input 1 (Component)	SOURCE 14\x0D
Input Input 1 (RGB)	SOURCE 11\x0D
Input LAN	SOURCE 53\x0D
Input Screen Mirroring	SOURCE 56\x0D
Input USB 1	SOURCE 52\x0D
Input USB 2	SOURCE 54\x0D
Input USB Display	SOURCE 51\x0D
Input Video (RCA)	SOURCE 41\x0D
Input Whiteboard	SOURCE 55\x0D
Lamp Mode Eco	LUMINANCE 01\x0D
Lamp Mode Normal	LUMINANCE 00\x0D
Menu Navigation Down	KEY 36\x0D
Menu Navigation Enter	KEY 16\x0D
Menu Navigation Escape	KEY 05\x0D
Menu Navigation Left	KEY 37\x0D
Menu Navigation Menu	KEY 03\x0D
Menu Navigation Right	KEY 38\x0D
Menu Navigation Up	KEY 35\x0D
Power Off	PWR OFF\x0D
Power On	PWR ON\x0D
Split Screen Off	SPS 01 00\x0D
Split Screen On	SPS 01 01\x0D
Split Screen Left Input HDMI 1	SPS 03 30\x0D

Global Scripter Module Communication Sheet

Revision: 1/30/2018

Split Screen Left Input HDMI 2	SPS 03 A0\x0D
Split Screen Left Input Input 1 (Component)	SPS 03 14\x0D
Split Screen Left Input Input 1 (RGB)	SPS 03 11\x0D
Split Screen Left Input LAN	SPS 03 53\x0D
Split Screen Left Input Screen Mirroring	SPS 03 56\x0D
Split Screen Left Input USB 1	SPS 03 52\x0D
Split Screen Left Input USB 2	SPS 03 54\x0D
Split Screen Left Input USB Display	SPS 03 51\x0D
Split Screen Left Input Video (RCA)	SPS 03 41\x0D
Split Screen Left Input Whiteboard	SPS 03 55\x0D
Split Screen Mode Size 1	SPS 02 00\x0D
Split Screen Mode Size 2	SPS 02 01\x0D
Split Screen Mode Size 3	SPS 02 02\x0D
Split Screen Right Input HDMI 1	SPS 04 30\x0D
Split Screen Right Input HDMI 2	SPS 04 A0\x0D
Split Screen Right Input Input 1 (Component)	SPS 04 14\x0D
Split Screen Right Input Input 1 (RGB)	SPS 04 11\x0D
Split Screen Right Input LAN	SPS 04 53\x0D
Split Screen Right Input Screen Mirroring	SPS 04 56\x0D
Split Screen Right Input USB 1	SPS 04 52\x0D
Split Screen Right Input USB 2	SPS 04 54\x0D
Split Screen Right Input USB Display	SPS 04 51\x0D
Split Screen Right Input Video (RCA)	SPS 04 41\x0D
Split Screen Right Input Whiteboard	SPS 04 55\x0D
Split Screen Swap None	SPS 05\x0D
Volume 0	VOL 0\x0D
Volume 20	VOL 243\x0D

Appendix B. Update Commands

AV Mute	MUTE?\x0D
Aspect Ratio	ASPECT?\x0D
Device Status	ERR?\x0D
Freeze	FREEZE?\x0D
Input	SOURCE?\x0D
Lamp Mode	LUMINANCE?\x0D
Lamp Usage	LAMP?\x0D
Volume	VOL?\x0D